THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 11: **Electricity** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The number of electrons passing per second through a conductor carrying 3.2 A is 1

(a) 2×10^{18} **(b)** 2×10^{19} **(c)** 5×10^{18} **(d)** 3.2×10^{19}

2. Three resistors of 6 ohm each are connected in parallel. The equivalent resistance is 1

(a) 18 ohm **(b)** 2 ohm **(c)** 3 ohm **(d)** 6 ohm

3. The power of an electric device that draws 5 A at 220 V is 1

(a) 1.1 W **(b)** 44 W **(c)** 225 W **(d)** 1100 W

4. A charge of 600 C flows in 5 minutes. The current is 1

(a) 2 A **(b)** 120 A **(c)** 3 A **(d)** 0.5 A

5. A 10 ohm and a 40 ohm resistor are in parallel. The equivalent resistance is 1

(a) 50 ohm **(b)** 8 ohm **(c)** 5 ohm **(d)** 25 ohm

6. A 100 W bulb has a resistance at 220 V of 1

(a) 100 ohm **(b)** 22 ohm **(c)** 484 ohm **(d)** 2.2 ohm

7. The work done in moving 4 C of charge across a potential difference of 9 V is 1

(a) 2.25 J **(b)** 36 J **(c)** 13 J **(d)** 5 J

8. Resistors of 2 ohm, 3 ohm and 6 ohm are connected in parallel. The equivalent resistance is 1

(a) 11 ohm **(b)** 3 ohm **(c)** 1 ohm **(d)** 2 ohm

9. A 2 kW heater runs for 30 minutes. Energy consumed is 1

(a) 2 kWh **(b)** 4 kWh **(c)** 1 kWh **(d)** 60 kWh

10. The V-I graph of a conductor has a slope of 5 ohm. If the current is 2 A, the voltage is 1

(a) 10 V **(b)** 0.4 V **(c)** 2.5 V **(d)** 7 V

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. How many electrons flow per second through a conductor carrying a current of 0.8 A? 2

12. A 20 ohm resistor is connected in parallel with a 5 ohm resistor. Find the equivalent resistance. 2

13. Why is the series arrangement not used for home appliances? 2

14. A 10 ohm resistor and a 15 ohm resistor are connected in series to a 5 V cell. Find the current. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. A battery of 9 V is connected to 3 ohm, 4 ohm and 2 ohm in series. Find the current and the voltage across the 4 ohm resistor. 3

16. Explain with the help of a formula why a thick wire is used for heavy current. 3

17. How many 220 ohm resistors must be connected in parallel to draw a total current of 5 A from a 220 V line? 3

OR

Find the equivalent resistance of 3 resistors of 4 ohm, 6 ohm and 12 ohm connected in parallel and the current drawn from 6 V. 3

18. Why do two bulbs of 60 W and 100 W, both rated 220 V, have different resistances? Which has more? 3

OR

Find the energy consumed by three bulbs of 60 W, each used for 4 h daily for 30 days. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. A circuit has three resistors: 6 ohm and 3 ohm in parallel, in series with 4 ohm, connected to a 12 V battery. Find 5
(a) the total resistance
(b) the current from the battery
(c) the voltage across the parallel combination
(d) the current through each parallel resistor.

OR

Explain Joule's law of heating. Find the heat produced when a current of 2 A flows through a 50 ohm resistor for 10 minutes. State the heat in calories (1 cal = 4.2 J). 5

20. Three resistors of 10 ohm, 20 ohm and 30 ohm are in series. 5
(a) Find the equivalent resistance.
(b) What is the current if a 12 V battery is used?
(c) Find the power dissipated in each.

OR

How can you use four (or fewer) resistors of 2 ohm each to get 5
(a) 8 ohm
(b) 0.5 ohm
(c) 2 ohm
(d) 3 ohm?

END OF QUESTION PAPER Total: 40 Marks